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Aquanutri: A Mobile Platform for Skin-Based Deficiency Detection and Nutrient-Rich Fish Recommendation

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Abstract

AQUANUTRI functions as a modern AI-powered system which examines skin images for nutritional deficiencies while promoting sustainable fish farming on terraces for dietary improvements. This system distinguishes visible skin symptoms related to vitamin and mineral deficiencies using a CNN named ResNet50 that operates on a DermNet dataset which received specialized training. The system references identified deficiencies to its fish database and suggests personalized recommendations based on user location together with environmental conditions ranging from freshwater to brackish to marine conditions. AQUANUTRI functions by referring suitable fish species which maintain nutritional richness for small aquaponic farming ac- tivities whereas the information point toward eco-friendly self- sufficiency options. AQUANUTRI emerges as a valuable solution for nutritional health management because it implements deep learning connectivity between expert-data and inexpensive agri- cultural recommendations.

Keywords: Nutritional Deficiency Detection, Skin Image Analysis, Convolutional Neural Networks (CNNs), ResNet50, DermNet Dataset, Sustainable Aquaculture, Fish Nutrient Rec- ommendation, Precision Nutrition, Mobile Health Applications (mHealth), Terrace Fish Farming, AI in Public Health.

1. INTRODUCTION

A. Background

Nutritional deficiencies remain a significant global health concern, particularly in regions with limited access to diversi- fied diets and healthcare resources. Many micronutrient imbal- ances, such as deficiencies in vitamins A, D, B12, and minerals like iron and zinc, manifest visibly on the skin in the formof discoloration, dryness, rashes, or scaling. These physical cues, often overlooked, provide a non-invasive window into an individual's nutritional health status.

With the rise of deep learning and computer vision, skin image analysis using Convolutional Neural Networks (CNNs) offers a promising path for automated detection of such conditions. ResNet50, a robust CNN architecture [7],[10], has shown strong performance in visual classification tasks and is well-suited for identifying subtle dermatological patterns. The DermNet dataset [8], comprising thousands of labeled skin condition images, serves as an effective training source for such models.

On the other hand, sustainable aquaculture—particularly small-scale terrace-based fish farming—has emerged as a viable method to combat food insecurity and malnutrition. Fish are nutrient-dense, especially rich in omega-3 fatty acids, protein, and key vitamins. Harnessing this nutritional value in a targeted, personalized manner can significantly improve dietary health in resource-constrained communities [1],[2],[3].

B. Motivation

This project, titled AQUANUTRI, is driven by the dual mo- tivation of leveraging AI for health diagnostics and promoting eco-friendly dietary interventions. Current healthcare models are often reactive and disconnected from agricultural or dietary solutions. AQUANUTRI bridges this gap by providing a proactive, data-driven system that detects potential nutritional deficiencies from skin images and then recommends locally appropriate fish species based on habitat compatibility and nutrient content.

Incorporating user-specific parameters like country and wa- ter habitat (freshwater, brackish, or marine), the system ensures contextual relevance in its fish nutrient recommendations. The end goal is to empower individuals and communities with an accessible mobile-based solution that not only informs but enables sustainable action through terrace aquaponics. AQUANUTRI aspires to contribute to public health by uniting AI diagnostics with nutritional ecology in a single, scalable platform

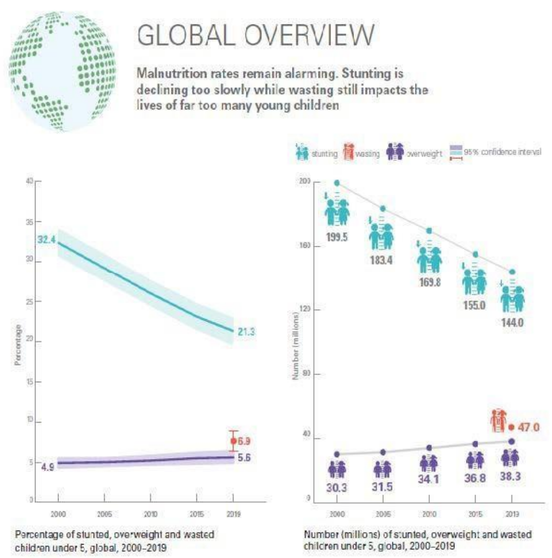

Fig. 1. Global Overview of Micronutrient Deficiency Patterns and Inequitable Access to Nutritional Resources Across Diverse Populations

2. LITERATURE REVIEW

The intersection of artificial intelligence, nutrition science, and sustainable aquaculture has garnered increasing attention due to its potential to address malnutrition in underserved populations.

A. AI-Based Detection of Nutritional Deficiencies

Numerous studies have explored the application of computer vision and deep learning for identifying nutritional deficiencies through visual analysis of dermal features. For instance, Smith and Jones [5] highlighted how micronutrient deficiencies such as low levels of iron, zinc, and vitamin B12 manifest in the skin through discoloration, pallor, or erythema. Sharma et al. [6] developed machine learning-based diagnostic tools for detecting such

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symptoms. Pioneering works by Litjenset al. [7] and Esteva et al. [8] demonstrated the efficacy of convolutional neural networks (CNNs) in dermatological image classification, achieving performance on par with clin- ical experts. Lee and Kwon [9] further confirmed the abilityof CNNs to detect nutrient-deficiency-related facial and skin features—providing a strong foundation for AQUANUTRI's use of ResNet50. Xu et al. [10] provided a comprehensive review of image recognition algorithms for medical diagnosis, validating the viability of AI-driven detection systems.

B. Nutritional Role of Fish in Diet

Fish are a vital source of essential nutrients such as omega-3 fatty acids, iodine, iron, and vitamin D. Tacon and Metian [3] emphasized the global importance of fish in addressing nu- tritional deficiencies. The USDA Nutrient Database [4] offers extensive data on the macro- and micronutrient composition of commonly consumed fish species. Studies by Bogard etal. [15] and Golden et al. [16] link regular fish consumptionto improved dietary quality, especially in low- and middle- income populations—supporting AQUANUTRI's fish-based dietary intervention strategy.

C. Terrace Aquaculture and Sustainable Food Systems

Urban and terrace-based aquaculture systems have emerged as promising solutions for localized, sustainable food pro- duction. Miao and Lal [1] discussed the challenges and opportunities of implementing terrace aquaculture in dense urban settings. Edwards [2] positioned urban aquaculture asa strategy to combat food insecurity. Additionally, Love et al. [17] and Maucieri et al. [18] explored the integration of aquaponics and aquaculture technologies for efficient, space- conserving food production—concepts that align closely with AQUANUTRI's sustainability goals.

D. Mobile Health Applications and Usability

Mobile health (mHealth) applications have revolutionized healthcare delivery by enabling remote diagnosis and per- sonalized interventions. Luxton et al. [11] reviewed the ef- fectiveness of mobile apps in diagnosing and managing mi- cronutrient deficiencies. Usability studies by Miller et al. [12] stressed the importance of intuitive interfaces and personalized dietary feedback. Visual and interactive data representation, as discussed in [13], [14], enhances user comprehension and trust in AI-driven recommendations—an essential aspect of AQUANUTRI's mobile-centric approach.

E. AI in Nutritional Recommendation Systems

Ng et al. [19] explored the use of machine learning algo- rithms to predict nutrient deficiencies based on dietary intake data. Such models support the personalized fish recommen- dation engine in AQUANUTRI, enabling dietary interventionsthat consider both real-time diagnostics and user-specific pref-erences.

Overview of Aquanutri

A. Application Purpose

AQUANUTRI is designed as an intelligent and accessible mobile health application that combines skin-based deficiencydetection with personalized dietary recommendations rooted insustainable aquaculture. The system identifies early signs of nutritional deficiencies using skin image analysis and, based on these findings, recommends fish species that can address thespecific nutrient gaps. By integrating AI-powered image recog- nition with a curated fish nutrient database, AQUANUTRI bridges the gap between diagnosis and dietary

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intervention. It further supports users by suggesting fish species compatible with their local habitat—freshwater, brackish, or marine—and promotes terrace fish farming as an eco-conscious, space- efficient solution to enhance nutritional intake.

B. Target Audience

The primary users of AQUANUTRI include individuals in underserved or rural communities with limited access to healthcare and nutrition guidance. It also caters to urban residents interested in home-based aquaculture as a sustainablefood source. Additionally, the application is intended for health workers, NGOs, and government programs focused on combating malnutrition. By delivering AI-driven diagnostics and practical recommendations through a user-friendly inter- face, AQUANUTRI aims to empower both individuals and communities with the tools necessary for self-monitoring and proactive health improvement.

Key Features

A. AI-Based Skin Image Analysis

AQUANUTRI utilizes Convolutional Neural Networks (CNNs), specifically the ResNet50 architecture, to analyze skin images with high precision. The model has been fine- tuned using a combination of ImageNet and DermNet datasets,enabling it to detect symptoms such as pale skin, discol- orations, rashes, and other visible signs associated with defi- ciencies in nutrients like iron, vitamin B12, and vitamin D. By leveraging mobile camera input, the system provides real-time diagnostic feedback, reducing the need for early-stage clinical testing and offering underprivileged communities access to essential preventive healthcare tools [5], [6], [10].

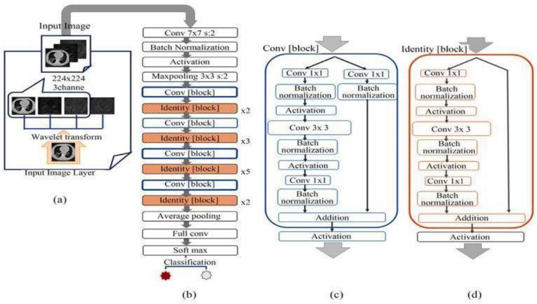

Fig. 2. Sequential Flow of Skin Image Processing Through Convolu- tional Neural Network Architecture Featuring Preprocessing Layers, Residual Blocks, and Final Classification Activation

B. Fish Nutrient Recommendation System

Following the identification of specific deficiencies, AQUANUTRI accesses a comprehensive nutrient database detailing the macro- and micronutrient composition of variousfish species. The system generates personalized fish recom- mendations based on the user’s geographic location and local aquatic environment. For example, if deficiencies in omega-3 or iodine are detected, species such as sardines or milkfish

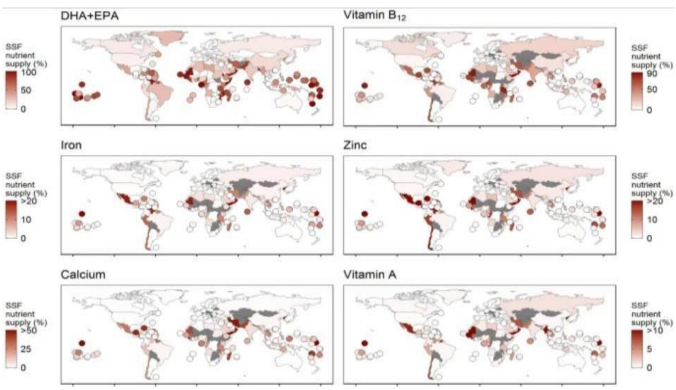

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Fig. 4. Contribution of Small-Scale Fish Farming (SSF) to Dietary Intakeof Essential

Micronutrients Including DHA, EPA, Vitamin B12, Iron, Zinc, Calcium, and Vitamin A Across Population Nutritional Requirements may be suggested [3],[15],[16]. This targeted recommendationsystem not only addresses health needs but also promotes the consumption of sustainable, locally-available protein sources

C. Support for Terrace-Based Aquaculture

As urban expansion limits traditional agricultural space, terrace and rooftop aquaculture systems present practical alter- natives. AQUANUTRI encourages the adoption of home-based terrace aquaponic systems for fish cultivation [1],[2],[18]. It provides users with essential guidance on selecting appropriate tank sizes, water types, feeding schedules, and species options suitable for closed or semi-open systems [3]. This feature supports both food self-sufficiency and environmental sustain- ability while contributing to global sustainable development goals.

D. Mobile-First Health Monitoring

Designed with accessibility in mind, AQUANUTRI offers a mobile-first interface suitable for daily use. Users can upload skin images, set preferences, and receive instant diagnostic feedback. Integrated widgets and filters allow precise trackingof nutrient levels, while visual tools enhance user understand- ing [11],[12]. The mobile format empowers community health workers to conduct screenings outside clinical settings, and the app's additional features—such as report generation and dietary plan customization—make it a comprehensive tool for both detection and long-term nutritional support.

Interactive Features

A. Real-Time Nutritional Deficiency Feedback

AQUANUTRI's diagnostic system provides immediate feedback following the upload of a skin image. The AI- based inference engine identifies visible skin indicators such as discoloration, dryness, and pigmentation irregularities that may signify deficiencies in vitamin B12, vitamin D, iron, and zinc. The results are accompanied by confidence scores to reflect model certainty and are highlighted using visual alerts with color-coded urgency indicators. The detection interface not only reveals the identified deficiencies but also includes concise descriptions of relevant nutrients and the health conse- quences of their deficiency [2], [17]. This real-time feedback mechanism helps users better understand their health status before seeking clinical attention, especially during rapid com- munity screening programs.

B. Customizable Fish Filter Options

AQUANUTRI’s recommendation engine allows users tofine-tune fish suggestions based on various parameters. Filtersinclude water type (freshwater, brackish, or marine), key nutrients (e.g., omega-3, iodine, protein), specific dietary pref- erences (e.g., vegetarian-friendly, low-fat), and market avail- ability. These options ensure that the recommendations are nutritionally effective, environmentally suitable, and culturally appropriate. Recommendations differ based on user location; for instance, inland populations receive different suggestions than coastal communities. The system also considers seasonal fish availability, supporting ethical harvesting practices while offering users a variety of diet options [3],[15],[16].

C. Visual Deficiency Mapping and Severity Insights

The application overlays visual markers and heat zones on the analyzed skin image to indicate regions that raised concern during diagnosis. Color-coded severity levels—from mild to moderate to severe—help users understand the seri- ousness of the detected symptoms. Interactive elements allow users to click on highlighted zones to access more detailed explanations. This visual and interactive feedback enhances user understanding of the AI’s findings and fosters trust in the system’s output. Additionally, it serves as a helpful visual tool for remote healthcare providers when presenting resultsto patients with limited medical literacy.

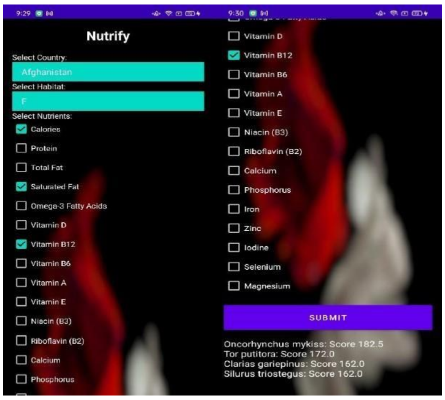

Fig. 5. Fish nutrient recommendation screen providing personalized sugges- tions based on deficiency results.

D. Progress Tracking and Personal Health Timeline

AQUANUTRI enables users to maintain a record of their skin assessments, generating a personalized health timeline that highlights changes in nutritional status over time. Each skin analysis contributes to a visual progression chart that reflects improvements, recurring issues, or new concerns. Users can set health goals and receive automatically gener- ated recommendations tailored to their progress. The time- line includes visually appealing graphs and summary cards that simplify pattern recognition. A continuous monitoring mechanism, supported by ExcelConnect, empowers users to track their nutritional changes and stay committed to long- term personal health management.

Implementation

A. System Architecture and Workflow

The AQUANUTRI system comprises a smartphone-friendly visual interface and a Flask-

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based API service, powered by a customized ResNet50 model trained on DermNet data. The CNN model processes skin images uploaded by users to detect signs of nutritional deficiencies during the preprocessing stage. Based on detected deficiencies, the system accesses a fishnutrient database and applies filters using user inputs such as water type and geographic location. Suitable fish species are then recommended. The results are displayed in a real-time, user-friendly format through the mobile interface, ensuring a seamless and interactive diagnostic experience.

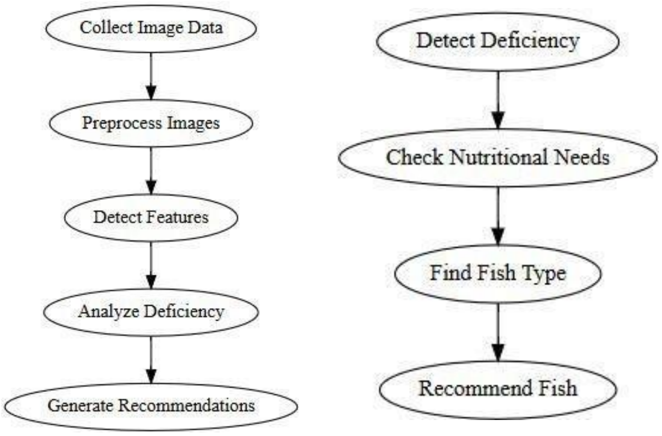

Fig. 6. Sequential Workflow of AQUANUTRI's Skin-Based Nutritional Deficiency Detection and Fish Recommendation Pipeline Integrating Image Processing, Feature Extraction, and Personalized Dietary Guidance

B. Model Training and Data Integration

The ResNet50 model employed in AQUANUTRI was fine- tuned using transfer learning on the DermNet dataset, with a specific focus on symptoms related to nutritional deficiencies. Data augmentation techniques were applied to enhance model generalization and robustness. The training pipeline included iterative validation and optimization steps to improve accuracy and reliability. A structured CSV-based database contains fish nutrient profiles, mapping specific deficiencies to appropriate fish sources [6],[9],[19]. The system leverages the pandas library for efficient data filtering and generates personalized dietary suggestions. The overall architecture is optimized for mobile deployment, delivering fast inference speeds and high responsiveness.

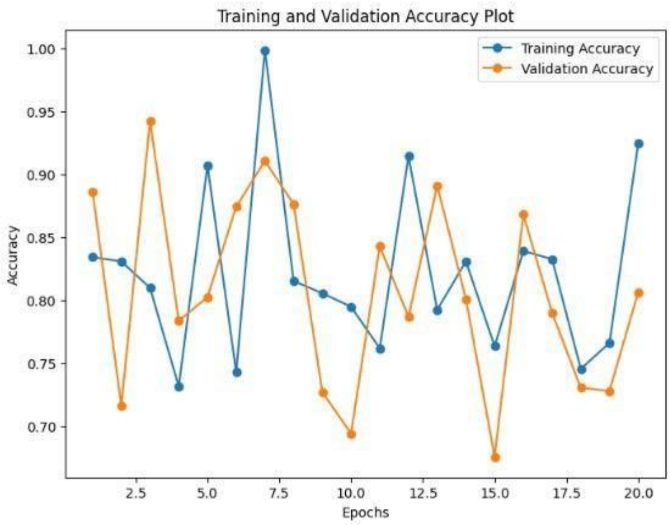

Fig. 7. Training and validation accuracy/loss curve during model develop- ment.

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Impact and Future Directions

A. Societal and Health Impact

1. AQUANUTRI provides everyone access to easy detec- tion tools for early nutritional deficiencies and promotes sustainable dietary solutions based on fish as resources.
2. The mobile artificial intelligence system enables health monitoring access to populations who lack proper health-care services.
3. The system promotes health-based diet planning which matches regional food accessibility.
4. Urban Food Security – Promotes terrace aquaculture, supporting self-reliant food production [16].
5. Environmental Sustainability – Reduces reliance on industrial food systems with eco-friendly practices.
6. Community Health Awareness – Fosters preventive health culture through visual insights and food education.

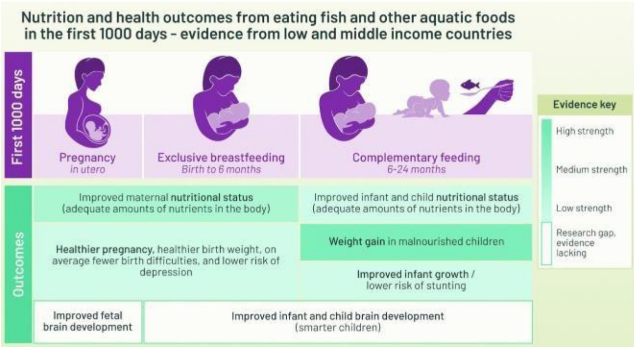

Fig. 8. Developmental and Nutritional Impacts of Aquatic Food Consumption During the First 1000 Days of Life in Low- and Middle-Income Contexts

B. Future Enhancements

1. System enhancements will enable smarter technology operation and enhanced data precision with better con- nection to health and farming network systems.
2. The AI model needs expansion to detect more than one nutrient deficiency at once for multi-nutrient detection purposes.
3. Expanded Fish Database – Include more regional fish species and their nutritional profiles.
4. The system will implement IoT sensor systems for col- lecting data about terrace aquaculture environment status.
5. Multilingual Support – Increase accessibility through language localization.
6. Health API Integration – Sync with wearable devices and health platforms for better tracking.

2. CONCLUSION

AQUANUTRI employs AI technology combined with aqua- culture practices to solve problems linked to nutritional defi- ciencies. The system leverages deep learning models including ResNet50 to analyze real-time skin images through which it detects nutrient-related health issues that manifest externally

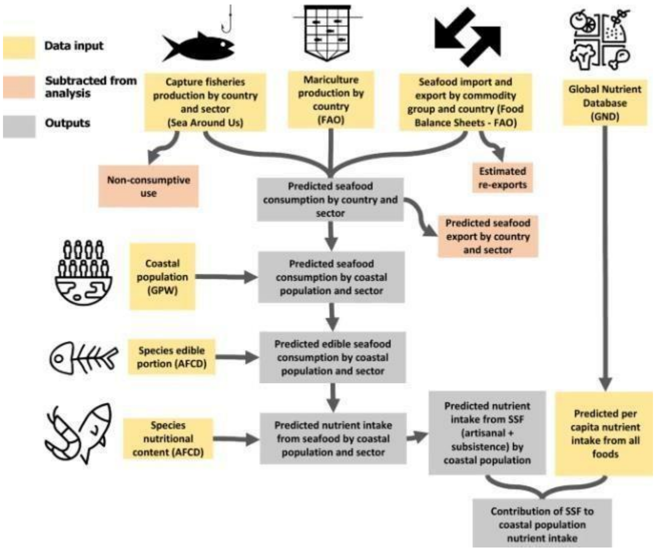

Fig. 9. Integrated Data Flow Model Illustrating Seafood Production, Trade, and Predicted

Nutritional Intake Among Coastal Populations Across Capture Fisheries and Mariculture Sectors in a user-friendly mobile application. The system provides more than diagnosis capabilities by delivering practical dietary suggestions for fish consumption which are designed to match user locations alongside personal preferences. Through its pro- motion of terrace-based aquaculture AQUANUTRI enhances food production practices using environmentally friendly methods as well as local food independence. AQUANUTRI operates as a complete health tracking device to transform public health nutrition through AI diagnostic systems com- bined with eco-friendly food solutions.

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